

DML Observation

Tables used: STUDENT, FACULTY, DEPT (renamed from DEPARTMENT), COURSE.

```
ALTER TABLE FACULTY ADD Age INT;
```

-- 1. INSERT: Populate all tables with data

```
INSERT INTO FACULTY (FacNo, FacName, Gender, DOB, DOJ, MobileNo, Dept, Age) VALUES
('F001', 'Ramesh', 'M', '1978-05-14', '2005-06-01', 9840012345, 'CSE', 48),
('F002', 'Mohan', 'M', '1965-11-02', '1998-07-15', 9840012346, 'CSE', 60),
('F003', 'Sunitha', 'F', '1980-03-22', '2008-08-20', 9840012347, 'ECE', 46),
('F004', 'Kavya', 'F', '1958-01-10', '1990-06-10', 9840012348, 'IT', 68),
('F005', 'Arjun', 'M', '1975-09-30', '2003-09-01', 9840012349, 'CSE', 51);
```

```
mysql> INSERT INTO FACULTY (FacNo, FacName, Gender, DOB, DOJ, MobileNo, Dept, Age) VALUES
-> ('F001', 'Ramesh', 'M', '1978-05-14', '2005-06-01', 9840012345, 'CSE', 48),
-> ('F002', 'Mohan', 'M', '1965-11-02', '1998-07-15', 9840012346, 'CSE', 60),
-> ('F003', 'Sunitha', 'F', '1980-03-22', '2008-08-20', 9840012347, 'ECE', 46),
-> ('F004', 'Kavya', 'F', '1958-01-10', '1990-06-10', 9840012348, 'IT', 68),
-> ('F005', 'Arjun', 'M', '1975-09-30', '2003-09-01', 9840012349, 'CSE', 51);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
INSERT INTO STUDENT VALUES
(191, 'Aravind', 'M', '2001-04-12', 9884012345, 'Chennai'),
(192, 'Divya', 'F', '2001-07-19', 9884012346, 'Madurai'),
(193, 'Karthik', 'M', '2000-12-05', 9884012347, 'Coimbatore'),
(194, 'Meena', 'F', '2001-02-27', 9884012348, 'Salem'),
(195, 'Suresh', 'M', '2000-08-16', 9884012349, 'Trichy');
```

```
mysql> INSERT INTO STUDENT VALUES
-> (191, 'Aravind', 'M', '2001-04-12', 9884012345, 'Chennai'),
-> (192, 'Divya', 'F', '2001-07-19', 9884012346, 'Madurai'),
-> (193, 'Karthik', 'M', '2000-12-05', 9884012347, 'Coimbatore'),
-> (194, 'Meena', 'F', '2001-02-27', 9884012348, 'Salem'),
-> (195, 'Suresh', 'M', '2000-08-16', 9884012349, 'Trichy');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

-- 2. SELECT: View all records from every table

```
SELECT * FROM STUDENT;  
SELECT * FROM FACULTY;  
SELECT * FROM DEPT;  
SELECT * FROM COURSE;
```

```
mysql> SELECT * FROM STUDENT;
```

RegNo	Name	Gender	DOB	MobileNo	City
191	Aravind	M	2001-04-12	9884012345	Chennai
192	Divya	F	2001-07-19	9884012346	Madurai
193	Karthik	M	2000-12-05	9884012347	Coimbatore
194	Meena	F	2001-02-27	9884012348	Salem
195	Suresh	M	2000-08-16	9884012349	Trichy

```
5 rows in set (0.02 sec)
```

```
mysql> SELECT * FROM FACULTY;
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	Dept	Age
F001	Ramesh	M	1978-05-14	2005-06-01	9840012345	CSE	48
F002	Mohan	M	1965-11-02	1998-07-15	9840012346	CSE	60
F003	Sunitha	F	1980-03-22	2008-08-20	9840012347	ECE	46
F004	Kavya	F	1958-01-10	1990-06-10	9840012348	IT	68
F005	Arjun	M	1975-09-30	2003-09-01	9840012349	CSE	51

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM DEPT;
```

DeptNo	DeptName	DeptHead
D01	CSE	F001
D02	ECE	F003
D03	IT	F004

```
3 rows in set (0.00 sec)
```

-- 2.1. Select Specific columns

```
SELECT Name, City FROM STUDENT;
```

```
mysql> SELECT Name, City FROM STUDENT;
+-----+-----+
| Name          | City          |
+-----+-----+
| Aravind Kumar | Chennai       |
| Divya         | Madurai       |
| Karthik       | Coimbatore    |
| Meena        | Salem        |
| Suresh        | Trichy        |
+-----+-----+
5 rows in set (0.00 sec)
```

-- 2.2. Select Using Where

```
SELECT * FROM FACULTY WHERE Dept = 'CSE';
```

```
mysql> SELECT * FROM FACULTY WHERE Dept = 'CSE';
+-----+-----+-----+-----+-----+-----+-----+-----+
| FacNo | FacName | Gender | DOB       | DOJ       | MobileNo | Dept | Age |
+-----+-----+-----+-----+-----+-----+-----+-----+
| F001  | Ramesh  | M      | 1978-05-14 | 2005-06-01 | 9840012345 | CSE  | 48  |
| F002  | Mohan   | M      | 1965-11-02 | 1998-07-15 | 9840012346 | CSE  | 59  |
| F005  | Arjun   | M      | 1975-09-30 | 2003-09-01 | 9840012349 | CSE  | 51  |
+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.02 sec)
```

-- 3. UPDATE: Modify existing records

```
-- Update the student name for a given register number
UPDATE STUDENT SET Name = 'Aravind Kumar' WHERE RegNo = 191;
```

```
mysql> UPDATE STUDENT SET Name = 'Aravind Kumar' WHERE RegNo = 191;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
```

```
-- Set age = 59 for faculty named 'Mohan'
UPDATE FACULTY SET Age = 59 WHERE FacName = 'Mohan';
```

```
mysql> UPDATE FACULTY SET Age = 59 WHERE FacName = 'Mohan';
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM FACULTY;
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	Dept	Age
F001	Ramesh	M	1978-05-14	2005-06-01	9840012345	CSE	48
F002	Mohan	M	1965-11-02	1998-07-15	9840012346	CSE	59
F003	Sunitha	F	1980-03-22	2008-08-20	9840012347	ECE	46
F004	Kavya	F	1958-01-10	1990-06-10	9840012348	IT	68
F005	Arjun	M	1975-09-30	2003-09-01	9840012349	CSE	51

```
5 rows in set (0.00 sec)
```

-- 4. DELETE: Remove records

```
-- Delete the faculty record who resigned (FacNo 'F004')
DELETE FROM FACULTY WHERE FacNo = 'F004';
```

```
mysql> DELETE FROM FACULTY WHERE FacNo = 'F004';
Query OK, 1 row affected (0.02 sec)
```

```
-- Remove all faculty over 65 years of age
DELETE FROM FACULTY WHERE Age > 65;
```

```
mysql> -- Remove all faculty over 65 years of age
mysql> DELETE FROM FACULTY WHERE Age > 65;
Query OK, 0 rows affected (0.02 sec)

mysql> |
```

```
SELECT * FROM STUDENT;
SELECT * FROM FACULTY;
```

```
mysql> SELECT * FROM STUDENT;
```

RegNo	Name	Gender	DOB	MobileNo	City
191	Aravind Kumar	M	2001-04-12	9884012345	Chennai
192	Divya	F	2001-07-19	9884012346	Madurai
193	Karthik	M	2000-12-05	9884012347	Coimbatore
194	Meena	F	2001-02-27	9884012348	Salem
195	Suresh	M	2000-08-16	9884012349	Trichy

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM FACULTY;
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	Dept	Age
F001	Ramesh	M	1978-05-14	2005-06-01	9840012345	CSE	48
F002	Mohan	M	1965-11-02	1998-07-15	9840012346	CSE	59
F003	Sunitha	F	1980-03-22	2008-08-20	9840012347	ECE	46
F005	Arjun	M	1975-09-30	2003-09-01	9840012349	CSE	51

```
4 rows in set (0.00 sec)
```